

CV MAKER

Android Application — Project Documentation

Technology: Kotlin • Jetpack Compose • Room Database • MVVM

Version: MVP / Version 1.0

Scope: One fixed professional CV template, multiple saved CVs, local-first storage, preview, PDF export and sharing.

Important MVP Constraint

AI integration is intentionally excluded from Version 1.0. Multiple templates, cloud synchronization, authentication and AI-assisted content generation are also outside the MVP.

1. Project Overview

CV Maker is a modern Android application that allows users to create, save, edit, preview, share and export professional single-page CVs. The application is local-first: CV data is persisted on the device using Room Database.

The MVP uses one fixed CV template. Users can create multiple CV records, but all records are rendered using the same professional template. The architecture is intentionally modular so future features such as AI assistance, additional templates and cloud synchronization can be introduced later.

2. Project Goals

- Provide a simple and professional CV creation experience.
- Allow users to create and manage multiple CVs.
- Persist CV information locally using Room Database.
- Allow users to edit and delete existing CVs.
- Provide a professional single-page CV preview.
- Export CVs as A4 PDF documents.
- Share generated CVs using Android's standard sharing mechanism.
- Allow basic appearance and CV font customization.
- Provide local data export and destructive-data controls.
- Use modern Android architecture and maintainable Kotlin code.
- Keep the architecture extensible for future AI integration.

3. MVP Scope

Included

- Splash screen
- Home dashboard
- Create CV
- Edit CV
- Save CV
- Multiple CV storage
- My CVs list
- CV preview
- Share CV
- Export CV as PDF
- Delete individual CV
- Theme selection
- Default font selection
- Export application data
- Delete all CVs
- About, Privacy Policy and Rate App

Excluded from MVP

- AI integration and AI content generation
- Multiple CV templates and template marketplace

- User authentication
- Cloud database and synchronization
- AI grammar correction, CV scoring or job-description optimization
- Subscriptions and premium features
- Online account-based backup

4. Technology Stack

Area	Technology / Approach
Language	Kotlin
UI	Jetpack Compose
Design System	Material 3
Architecture	MVVM + Repository
Local Storage	Room Database / SQLite
State	StateFlow + Compose state
Navigation	Navigation Compose
Async Work	Kotlin Coroutines
PDF	Android PDF APIs / document generation
Sharing	Android Sharesheet / Intent
Build	Gradle
DI	Manual initially; Hilt can be introduced later

5. Functional Requirements

Functional requirements describe what the application must do from the user's perspective.

FR-01 — Application Launch

- Display a branded splash screen.
- Navigate from Splash to Home after initialization.
- Do not place onboarding or CV-entry forms on the splash screen.

FR-02 — Home Dashboard

- Provide Create New CV, My CVs and Settings.
- Display recent CVs when available.
- Display an empty state when no CV exists.
- Use a Floating Action Button for the main actions.

FR-03 — Create CV

- Create a new CV.
- Provide a vertically scrollable Compose editor.
- Collect personal, contact, summary, education, experience, skills, projects, certifications and languages.
- Allow a profile photo.

- Save locally and preview the CV.

FR-04 — Edit CV

- Select an existing CV.
- Load its data from Room.
- Modify CV fields.
- Persist changes when Save is selected.

FR-05 — My CVs

- Display all saved CVs.
- Show identifying information such as name/title and last-updated date.
- Open preview.
- Edit, share, export or delete a CV.
- Provide search if enabled in final UI.

FR-06 — CV Preview

- Render data using the single fixed template.
- Display an A4-like document.
- Provide Edit, Share and Export PDF actions.
- Keep preview and PDF visually consistent.

FR-07 — PDF Export

- Export a CV as PDF.
- Use A4 dimensions.
- Produce a single-page document for the MVP.
- Include saved CV information.
- Respect the selected font where technically supported.

FR-08 — Share CV

- Share the generated CV.
- Use Android's standard Sharesheet.
- Expose the PDF through a shareable content URI.

FR-09 — Delete CV

- Delete an individual CV.
- Require confirmation.
- Remove the record from Room and My CVs.

FR-10 — Settings

- Choose System, Light or Dark theme.
- Choose default CV font.
- Open About, Privacy Policy and Rate App.
- Export local data.
- Delete all CVs after confirmation.
- Do not show Default Template because only one template exists.

FR-11 — Data Export

- Export stored CV information.
- Use JSON or another structured format for MVP.
- Include data required to reconstruct saved CV records.

FR-12 — Delete All Data

- Provide Delete All CVs.
- Show a destructive warning.
- Remove all CV records after confirmation.

6. Non-Functional Requirements

NFR-01 — Performance

- Common interactions should feel immediate.
- Do not block the main thread with Room or PDF work.
- Maintain smooth scrolling.
- Avoid unnecessary recompositions and expensive work in composables.

NFR-02 — Reliability

- Saved data should survive normal app restarts.
- Handle database failures gracefully.
- Do not lose saved data because of normal navigation.
- Require confirmation for destructive actions.

NFR-03 — Usability

- Keep the UI understandable for first-time users.
- Make Create, Save, Preview, Share and Export easy to find.
- Use clear labels and suitable keyboard/input types.
- Provide actionable validation messages.

NFR-04 — Accessibility

- Use meaningful content descriptions where needed.
- Maintain sufficient contrast.
- Use appropriate touch target sizes.
- Support system font scaling where practical.
- Do not communicate information by color alone.

NFR-05 — Maintainability

- Separate UI, ViewModel, repository and database responsibilities.
- Do not put business logic directly in composables.
- Extract reusable Compose components.
- Keep database access behind DAO/repository layers.
- Follow consistent Kotlin naming and style.

NFR-06 — Extensibility

- Permit future multiple templates.
- Permit future AI services.

- Allow repeated CV sections to become separate Room entities.
- Allow future cloud synchronization without replacing the Compose layer.

NFR-07 — Offline Capability

- Core creation, editing, viewing and storage work without internet.
- Room is the local source of truth in MVP.

NFR-08 — Security and Privacy

- Keep CV data local by default.
- Do not transmit CV data to a server in MVP.
- Handle generated files with appropriate Android URI/storage mechanisms.
- Request only necessary permissions.

NFR-09 — Compatibility

- Target a current stable Android SDK.
- Choose minimum SDK based on required Compose, Room and PDF APIs.
- Adapt UI to common phone screen sizes.

NFR-10 — Visual Quality

- Use consistent Material 3 styling.
- Make the CV look like a professional A4 document.
- Keep PDF output close to preview.
- Maintain consistent spacing, typography, colors and shapes.

7. Screen Specifications

Screen	Purpose	Main Actions
Splash	Branding and initialization	Navigate to Home
Home	Dashboard and quick actions	Create, My CVs, Settings
Create/Edit CV	Enter or modify CV data	Save, Preview
My CVs	Manage saved CVs	Preview, Edit, Share, PDF, Delete
CV Preview	Display final single-page CV	Edit, Share, Export PDF
Settings	Application preferences	Theme, Font, Data, App
Theme	Appearance selection	System, Light, Dark
Font	Default CV typography	Select font

8. CV Data Model and Room Database

Group	Fields
Identity	id, cvName
Personal	fullName, professionalTitle, profileImage
Contact	email, phone, location, linkedin, website

Group	Fields
Summary	summary
Education	degree, institution, start/end date, description
Experience	jobTitle, company, location, start/end date, description
Skills	skills
Projects	projectName, technologies, description, projectUrl
Certifications	name, organization, year
Languages	languages
Preferences	font
Metadata	createdAt, updatedAt

MVP note: a single entity is acceptable for a first working version. A future version should consider separate one-to-many entities for Education, Experience, Project, Certification, Skill and Language so users can add unlimited entries.

9. Architecture and Data Flow

Compose UI → ViewModel → Repository → DAO → Room Database

- Compose displays UI and collects actions.
- ViewModel owns screen state and coordinates operations.
- Repository abstracts CV data access.
- DAO contains database queries.
- Room persists data in SQLite.

Reactive data should preferably be exposed as Flow from DAO and collected using lifecycle-aware patterns.

10. Suggested Project Structure

```
com.example.cvmaker ├── data | ├── local | | ├── AppDatabase.kt | | ├── CvDao.kt | | └── CvEntity.kt | └── repository |
└── CvRepository.kt ├── navigation | └── AppNavigation.kt ├── ui | ├── screens | | ├── splash | | ├── home | | ├── cv | |
└── mycvs | | ├── settings | ├── components | └── theme ├── viewmodel | ├── CvViewModel.kt | ├── HomeViewModel.kt |
└── SettingsViewModel.kt ├── pdf | └── CvPdfGenerator.kt └── MainActivity.kt
```

11. Navigation Specification

- Splash
- Home
- CreateCV
- EditCV/{cvId}
- MyCVs
- Preview/{cvId}
- Settings
- Theme
- Font

CreateCV and EditCV should share the same editor UI where practical. The difference is whether a new record is being created or an existing record is loaded by ID.

12. PDF Export and Sharing

- Load CV data from ViewModel/repository.
- Generate an A4 PDF using a dedicated generator.
- Render the fixed CV template.
- Keep the PDF single-page for MVP.
- Expose the file through a content URI.
- Use Android's standard Sharesheet.
- Share template/layout rules between Preview and PDF generation.

13. Validation and Error Handling

- Validate required fields before saving where defined by product design.
- Use suitable input types and basic email validation.
- Show clear inline errors or Snackbar messages.
- Handle database failures at the appropriate layer.
- Show an actionable error if PDF generation fails.
- Require confirmation before destructive operations.

14. Privacy and Security

- The MVP is local-first and does not require an account.
- CV data stays on-device unless the user explicitly shares/exports it.
- No AI provider receives CV data in MVP.
- No cloud synchronization is required for Version 1.0.
- Use Android content URIs rather than exposing private filesystem paths.

15. UI / UX Design Guidelines

- Premium, minimal and professional Android UI.
- Material 3 components with a clean white/light-gray background.
- Primary color may use indigo/blue around #4F46E5.
- Use Inter, Roboto or Poppins.
- Use approximately 16–20dp rounded cards and generous spacing.
- Avoid excessive decoration; the CV document is the visual focus.
- Provide clear empty states.
- Use confirmation dialogs for destructive actions.
- Do not introduce template galleries or AI UI in MVP.

16. Testing Requirements

Unit Testing

- Test ViewModel state transitions.

- Test CV validation.
- Test repository operations.
- Test formatting/PDF preparation logic.

Database Testing

- Test insert, update, query and delete.
- Test delete-all.
- Test persistence and data retrieval.

UI Testing

- Verify navigation.
- Verify create/edit flows.
- Verify empty/populated My CV states.
- Verify Settings.
- Verify confirmation dialogs.

PDF Testing

- Verify A4 dimensions.
- Verify one-page output for supported content limits.
- Verify important fields appear.
- Compare preview and PDF visually.

17. Development Phases

Phase	Deliverables
1. Setup	Compose, Material 3, project structure, theme, Gradle configuration
2. UI	Splash, Home, CV Editor, My CVs, Preview, Settings
3. Room	Entity, DAO, Database, Repository, ViewModel integration
4. CV Management	Create, Save, Load, Edit, Delete
5. PDF	A4 generator, preview consistency, file sharing
6. Settings	Theme, Font, Export Data, Delete All, About/Privacy/Rate
7. Testing	Unit, database, UI and PDF verification
8. Future	AI, multiple templates, cloud sync, accounts

18. MVP Definition of Done

- User can launch and reach Home.
- User can create and save a CV.
- Saved CVs survive normal application restarts.
- User can view all saved CVs.
- User can preview, edit and delete a CV.
- User can export a single-page A4 PDF.
- User can share the CV PDF.

- User can change theme and default font.
- User can export local CV data.
- User can delete all CVs after confirmation.
- AI integration is not required for MVP completion.
- Multiple templates are not required for MVP completion.

19. Future Roadmap

Version	Potential Features
1.1	Multiple education/experience/project/certification entries
1.2	Multiple professional CV templates
1.3	Cloud backup and synchronization
2.0	AI summary and experience assistance
2.1	AI CV scoring and job-description optimization

20. Key Technical Risks and Mitigations

Risk	Mitigation
CV exceeds one page	Define content limits, typography and spacing rules; validate PDF output.
Preview differs from PDF	Centralize template rules and test representative CVs.
Large form is hard to maintain	Use reusable Compose sections and clear state models.
Room model becomes complex	Normalize repeated sections when requirements expand.
Data loss during editing	Keep draft state in ViewModel and save explicitly.
File sharing errors	Use FileProvider/content URIs and Android Sharesheet.
Future AI creates coupling	Keep AI as a separate service/repository layer when introduced.

21. Final Product Vision

The MVP should provide a focused workflow: launch the app, create a professional CV, save it locally, preview it as a single-page document, edit it whenever necessary, export it as an A4 PDF and share it. The product should remain intentionally simple while using a clean architecture that can support future expansion.

Core principle: Build a reliable CV workflow first. Add AI, multiple templates, cloud services and other advanced capabilities only after the local-first MVP is stable.